Biplab Mandal

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Professional Summary

AI Engineer with production experience building **distributed, fault-tolerant microservices** and agentic AI services on **GCP**. Skilled in **Python, REST APIs**, and messaging/caching systems (**Redis, Celery**), with hands-on work in system-level testing, **CI-integrated automated testing**, service reliability, and applied networking (**HTTP/HTTPS, WebSockets, TCP/IP**, streaming APIs). Strong foundation in data structures, algorithms, and scalable platform design.

Education

- **Indian Institute of Information Technology Kottayam, Kerala** *Oct 2022 Apr 2026*
  B.Tech Computer Science Engineering

Experience

•Agrati Tech Labs *Apr 2026 - Present*
AI Engineer

- Developed a **Marketplace** search feature for the **Nutri AI health-tech** application using FastAPI, React Native/Expo, and SerpApi Google Shopping.
- Implemented natural-language product search with price filters, product validation, normalization, deduplication, Redis caching, and rate limiting.
- Built a user-approval flow displaying verified product details, images, prices, merchant links, and redirecting users to the selected product page.

•Mondee Tech Pvt. Ltd *Sept 2025 - Mar 2026*
Software Engineer Intern

- Delivered production-grade backend services under **SDLC** best practices for a **distributed, multi-tenant** travel platform; drove code documentation, peer review, and version-controlled deployment across cross-functional teams, shipping fault-tolerant **microservices** on **GCP** with zero missed release cycles.
- Designed and deployed **distributed AI agent services** (Price Recommendation, Audio Transcription) on **GCP** using **Google ADK**, implementing a dual-layer fallback architecture across **PostgreSQL** and external **HTTP APIs**, and exposed them via streaming **FastAPI** services handling real-time request/response cycles.
- Identified unacceptable hallucination rates in early agent outputs; redesigned agent instructions and guardrails to enforce grounding against retrieved source data, **cutting hallucinated outputs by 70%+** and restoring team trust in automated responses.
- Kept the platform online amid unreliable upstream data sources by engineering dual-layer fallback logic and fault-tolerant deployment design across **distributed service boundaries**, **sustaining 7+ months of uninterrupted production service continuity**.
- Authored system architecture documentation and designed test cases for **system-level and integration testing** ahead of every deployment, meeting internal audit and **QA** standards with zero post-release compliance flags.
- Integrated unit and integration test suites into a **Jenkins CI** pipeline triggered on every pull request and push, catching regressions before merge and enforcing test coverage ahead of each release cycle.

Technical Skills

Languages: C/C++, Python, Java, JavaScript, HTML5, CSS3
Backend & Distributed Systems: REST APIs, Microservices, FastAPI, Django, Node.js, Express, WebSockets, HTTP/HTTPS
Databases & Messaging: MongoDB, MySQL, Redis, working knowledge of RabbitMQ, ChromaDB (Vector Database)
Cloud & DevOps: GCP, Microsoft Azure, Docker, Kubernetes, Jenkins, Terraform, CI/CD, Git, GitHub, Linux (Ubuntu)
AI/ML & Agentic Frameworks: LLMs, RAG, Google ADK, LangChain, LangGraph, AI Agents, Model Context Protocol (MCP), Prompt Engineering, working knowledge of Vertex AI, Databricks, MLflow
Relevant Coursework: Data Structures & Algorithms (Java), Operating Systems, DBMS, Computer Networks (TCP/IP, congestion control)

Projects

- **CollabSync** | *Node.js, TypeScript, Express, Socket.IO, MongoDB, MySQL, JWT* *Nov 2024*
  - Built a real-time, **distributed** collaboration backend with **JWT-secured WebSockets over TCP**, achieving multi-user document sync at under **200ms** latency across concurrent client connections.
  - Designed a conflict-aware sync engine using operation-based updates and version tracking to resolve concurrent edit conflicts, eliminating data loss in distributed multi-user sessions.
  - Split persistence across **MongoDB** (documents, operation history) and **MySQL** (identity, permissions) for durable state recovery after server restarts, enabling clean recovery with no manual intervention.
  - Implemented hash maps for **O(1)** session lookups and queue-based operation ordering, **cutting sync latency by 40%** under load.
- **AI ComplianceBot** | *Django, Celery, Redis, LangChain, ChromaDB, Next.js, Docker, Kubernetes* *Dec 2025*
  - Built a **RAG**-based compliance assistant that grounds every response in retrieved regulatory source clauses, eliminating unverifiable answers over dense regulatory **PDFs**.

- Replaced slow synchronous document processing with an async, **message-queue-based** ingestion pipeline using **Django, Celery, and Redis**, decoupling processing from the request cycle and **cutting API response latency by 65%**.
- Built a semantic retrieval pipeline on **ChromaDB** with **SentenceTransformer** embeddings, achieving sub-second query latency at scale.
- Containerized the full system with **Docker** Compose, significantly reducing environment setup time for new contributors.
- Deployed the containerized stack to a local **Kubernetes** cluster (**minikube**) as separate Deployments per service (Django API, Celery worker, Redis, Next.js frontend) with **Horizontal Pod Autoscaling** configured to handle variable ingestion load.